Rewarding Learning

ADVANCED SUBSIDIARY (AS)
General Certificate of Education
2026

Digital Technology

Assessment Unit AS 1

assessing

Approaches to System Development

[SDT11]

FRIDAY 15 MAY, AFTERNOON

MARK SCHEME

General Marking Instructions

Introduction

Mark schemes are published to assist teachers and students in their preparation for examinations. Through the mark schemes teachers and students will be able to see what examiners are looking for in response to questions and exactly where the marks have been awarded. The publishing of the mark schemes may help to show that examiners are not concerned about finding out what a student does not know but rather with rewarding students for what they do know.

The Purpose of Mark Schemes

Examination papers are set and revised by teams of examiners and revisers appointed by the Council. The teams of examiners and revisers include experienced teachers who are familiar with the level and standards expected of students in schools and colleges.

The job of the examiners is to set the questions and the mark schemes; and the job of the revisers is to review the questions and mark schemes commenting on a large range of issues about which they must be satisfied before the question papers and mark schemes are finalised.

The questions and the mark schemes are developed in association with each other so that the issues of differentiation and positive achievement can be addressed right from the start. Mark schemes, therefore, are regarded as part of an integral process which begins with the setting of questions and ends with the marking of the examination.

The main purpose of the mark scheme is to provide a uniform basis for the marking process so that all the markers are following exactly the same instructions and making the same judgements in so far as this is possible. Before marking begins a standardising meeting is held where all the markers are briefed using the mark scheme and samples of the students' work in the form of scripts. Consideration is also given at this stage to any comments on the operational papers received from teachers and their organisations. During this meeting, and up to and including the end of the marking, there is provision for amendments to be made to the mark scheme. What is published represents this final form of the mark scheme.

It is important to recognise that in some cases there may well be other correct responses which are equally acceptable to those published: the mark scheme can only cover those responses which emerged in the examination. There may also be instances where certain judgements may have to be left to the experience of the examiner, for example, where there is no absolute correct response – all teachers will be familiar with making such judgements.

Candidates are expected to use Digital Technology terminology, appropriately and accurately, in their responses.

In questions asking candidates to 'identify' or 'list' or equivalent, responses consisting of appropriate **single words or short phrases are acceptable**. In all other questions, where the key word is 'explain' or 'describe' or 'evaluate' or equivalent, single words or short phrases are generally not acceptable; the candidate's presentation, spelling, punctuation and grammar should be clear.

Listing

Where **one** response is required to gain a mark, candidates will not gain credit if a correct response is given alongside one or more incorrect responses. This is referred to as listing.

AVAILABLE MARKS

1 (a) (i) To analyse the user requirements/data processing requirements
To conduct a feasibility study
To identify possible solutions
To identify the most suitable solution
To draw up the system specification
To respond to faults discovered during system testing
To liaise with client
(4 × [1]) [4]

(ii) **Note** *Advantages of interviews **not** disadvantages of questionnaires*
Advantage [1] Feature [1]
Interviews can be unstructured
… allowing follow up questions/clarification of questions/unexpected responses
(2 × [1])

As they are conducted on a one-to-one basis/in small groups
… users can express opinions in more detail/in their own words/ through non-verbal cues
(2 × [1])

Interviews have a higher rate of responses
… as people frequently ignore questionnaires
(2 × [1])
(2 × [2]) [4]

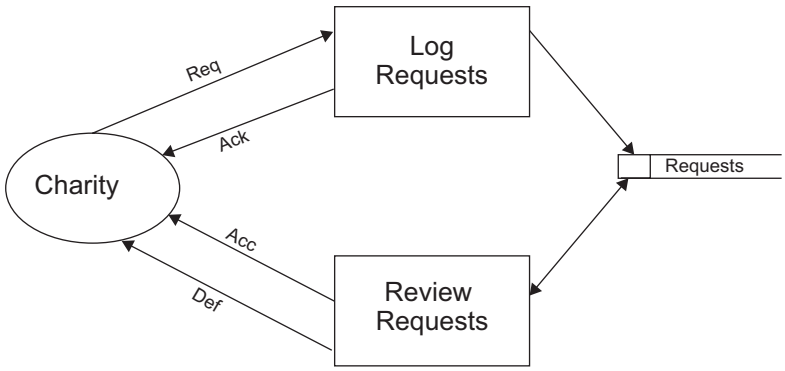
(iii) **Note** *Disadvantages of interviews **not** advantages of questionnaires*
Disadvantage [1] Feature [1]
Because they are conducted one to one/in small groups
… users may provide answers to suit the interviewer/may be reluctant to answer truthfully
(2 × [1])

Interviewing is more time consuming
… to plan and carry out/to summarise and analyse results digitally
(2 × [1]) [4]

(b) (i) A context level DFD consists of a single process
A level 1 DFD shows the main subprocesses
(2 × [1])

A level 1 DFD shows datastores
... and data flows
(2 × [1]) [4]

(ii)



[1] for external entity
[1] for each of two subprocesses **Log** and **Review** (Given in question)
[1] for datastore **Requests** (Given in question)
[1] for each of three internal dataflows
[1] for all four external dataflows
(8 × [1]) [8]

AVAILABLE MARKS

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2 (a) (i)

Reference	Component
A	Node number
B	Task identifier
C	Earliest start time [1]
D	Latest finish time [1]
E	Elapsed time of task [1]

[3]

(ii) Note *'how the critical path assists in the management of…'*
The critical path shows the sequence in which activities must be carried out
… to ensure the project is completed in the minimum time
(2 × [1])
Some project activities must be completed in a certain order/in parallel
Some activities depend on other activities/cannot start until other activities are completed
(2 × [1])
The project duration/progress to be determined
… enabling precise scheduling/cost control/allocation of resources
Bottlenecks/slowdowns can be identified/minimised
(3 × [1]) max [5]

(b) Constraint
A restriction
… which is placed on/acts on the project.
[1] + [1]
Example, e.g. scope [1]

Risk
A factor/consideration
… which might impact on the success of the project
[1] + [1]
Example, e.g. insufficient development resources [1]
(2 × [3]) [6]

AVAILABLE MARKS

(c) Note *'version management of source code during system development'*
Candidates must explicitly refer to source code
All changes made to the source code must be tracked
… and the reason for the changes recorded
… and who made the changes identified
Each version of code/program/software will be given a version number
All programmers must be working on the same version of the code
Changes to code can be reversed
(4 × [1]) [4]

(d) **(i)** To identify additions/modifications to data objects
… and how data objects are retrieved
(2 × [1]) [2]

(ii) **Banded response**

Features of RAD
Incremental model/iterative approach
The project is subdivided into a series of builds, each of which undergoes its own separate lifecycle, passing through requirements, design implementation and testing phases
Each component of the system is developed in parallel, within a given time frame and then assembled to produce a working prototype

Features of the waterfall model
Discrete, linear, self-contained stages
Deliverables
Repeating a stage/re-visiting an earlier stage

Comparison
RAD Continuous, active involvement of end user groups
Waterfall model End users only involved in fact finding and user testing

Level	Marking criteria	Marks
Band 2	The candidate • Provides a detailed and accurate description of each method which is correct • Compares both methods wrt time end user involvement • Uses the appropriate Digital Technology terminology accurately throughout the response Presentation, spelling, punctuation and grammar are of a high standard.	[5]–[6]
Band 1	The candidate • Provides a detailed and accurate description of each method • Discusses user involvement considerations • Uses some relevant Digital Technology terminology Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[3]–[4]
Band 0	The candidate • Provides a description of each method which is correct but lacks detail **or** makes a simple comparison of the users involvement in both methods • Makes limited use of Digital Technology terminology Presentation, spelling, punctuation and grammar are such that the intended meaning is not completely clear.	[1]–[2]

[6]

AVAILABLE MARKS

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3 (a) (i) Test reference/number
Purpose of test
Test data/Test data type
Expected result
Actual result
Tester ID
(4 × [1]) [4]

(ii) Extreme data
Data on the limits of acceptability [1]
Example 1000 [1]

Exceptional data
Invalid data which will not be accepted [1]
Example 2000 [1] [4]

				AVAILABLE MARKS
	(b)	**System testing** To test that each individual component works correctly … and that the system meets the system specification Performed in-house/by the development team ... using the test plan/schedule Comprises system testing/module or unit testing/integration testing White box/black box testing may be used (2 × [1]) **Acceptance testing** To allow a selection of end users … to test that the system meets the user requirements To test that the system meets the user requirements … in a live environment … with real volumes of data Leads to sign off (2 × [1])	[4]	12
4	**(a)**	Old database structures … converted to new database structure Data from old system … transferred to new system (4 × [1])	[4]	
	(b)	**(i)** The potential causes of the disaster … natural/man-made or example [1] + [1] The support needed to implement the plan … the technology/other resources [1] + [1] The scope of the plan … the areas of the organisation covered by the plan [1] + [1] Assumptions being made about the availability of resources during the disaster … such as technology/human resources [1] + [1] Key personnel … and their responsibilities [1] + [1] Key/critical data … purpose/location [1] + [1] An alternative location where the organisation can function … until the disaster is over [1] + [1] How the plan will be revised/tested … how frequently [1] + [1] Backup method … and recovery method [1] + [1] (3 × [2])	[6]	

AVAILABLE MARKS

(ii) Banded response

Incremental backup
This will make a copy of those data files that have changed since the last backup which may have been a full backup or an incremental backup
Full backup
This will make a copy of all data files and provides a complete snapshot of data at time of backup
Description of data recovery
Firstly, data from the previous week's full backup will be restored. Then, the daily incremental backups since the last full backup will be restored/ applied to the recovered data in reverse order to the order in which the incremental backups were performed

Level	Marking criteria	Marks
Band 2	The candidate • Provides a detailed description of each backup method which is correct • Provides a detailed description of the recovery method which is correct • Uses the appropriate Digital Technology terminology accurately throughout the response Presentation, spelling, punctuation and grammar are of a high standard.	[5]–[6]
Band 1	The candidate • Provides a detailed description of each backup method • Discusses user recovery • Uses some relevant Digital Technology terminology Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[3]–[4]
Band 0	The candidate • Provides a description of each backup method which lacks detail • Makes limited use of Digital Technology terminology Presentation, spelling, punctuation and grammar are such that the intended meaning is not completely clear.	[1]–[2]

[6] 16

5 (a) (i) Project file management support/directory trees
Design tools/GUI builder
A code editor/text editor
Smart editor/syntax checker/syntax suggestions/automatic tabs/ automatic comments/automatic white space/automatic line numbering
Code linking/execution
Debugging tools/breakpoints/variable inspection
(4 × [1]) [4]

		AVAILABLE MARKS
(ii)	An interpreter translates source code one line at a time … while a compiler converts the whole program into machine code [1] + [1] An interpreter executes one line of code at a time With compilation, the whole program is executed at once [1] + [1] A compiler produces object code/an executable version An interpreter does not produce object code [1] + [1] An interpreter can execute a partially completed/partially correct program With compilation, the program must be free of syntax errors before it can be executed [1] + [1] If there is an error during translation, the process stops With compilation, a list of errors is created [1] + [1] (2 × [2]) [4]	

(b)

```
IF      mark < 40
THEN
        grade = 'Fail'
ELSE IF     mark < 70
        THEN
                grade = 'Pass'
        ELSE
                grade = 'Distinction'
        END IF
```

Outer **IF THEN ELSE END IF** [4]
Nested **IF THEN ELSE END IF** [4]
+ [1] if all conditions/actions for outer selection are correct
+ [1] if all conditions/actions for inner selection are correct
(10 × [1]) [10]

(c) **Note** Candidates must use appropriate OO terminology

Object
An element of a program that can perform actions
… and interact with other elements of the program
(2 × [1])
Alternative
An object describes a real world element
… through the use of methods and properties
(2 × [1])

Method
A procedure contained within a class
… which defines an action that an object of that class can perform
(2 × [1])
Alternative
A section of code/procedure contained within a class
… which defines an action that an object of that class can perform.
(2 × [1])
(2 × [2]) [4] 22

Total **100**